HW 5 AND 6 FALL 2026
MATH 394 GEOM ADV VIEWPOINT I

This homework is due on the wednesday class, Nov 11, 2026.
Q5&6.1 [Lee, p.101, 4A]. Prove Theorem 4.9 (one ray must be between the other two).

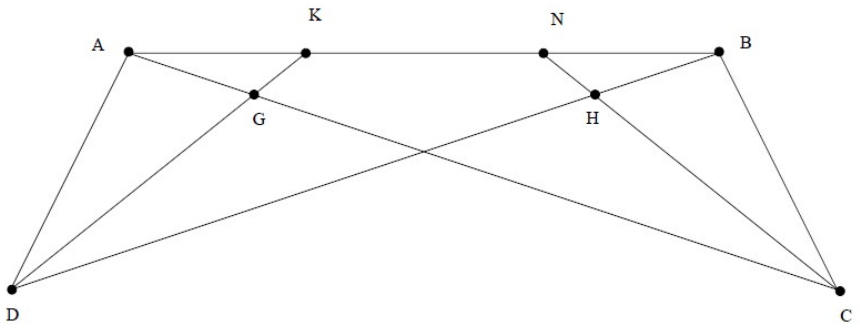
Q5&6.2. [Lee, p.101, 4C]. Prove Theorem 4.11 (Euclid's common notions for angles).

Q5&6.3. [Lee, p.101, 4G] Suppose $\vec{a}$, $\vec{b}$, $\vec{c}$, and $\vec{d}$ are four distinct rays with the same endpoint, such that $\vec{a}$ and $\vec{d}$ are opposite rays and $\vec{a} * \vec{b} * \vec{c}$. Show that $\angle ab$, $\angle bc$, and $\angle cd$ form a linear triple. (In other words, condition (iii) in the definition of a linear triple is redundant.)
(Hint: Use the interior lemma.)

Q5&6.4. [Lee, p.101, 4D]. Prove Theorem 4.17 (the vertical angles theorem).

Q5&6.5. [Lee, p.101, 4I]. Prove Theorem 4.30 (construction of perpendiculars).

Q5&6.6. [Use Basic Facts] In the figure below we assume $AD = BC$, $AC = BD$, $AK = BN$. Prove $KG = NH$.



Date: November 01, 2026.

Q5&6.7. [Use Basic Facts] Let $ABCD$ be a quadrilateral such that AB is parallel to CD (i.e. trapezoid, see figure below). Let E be the midpoint of AD and let F be the point of BC such that EF is parallel to AB. Prove that $EF = \frac{1}{2}(AB + CD)$.
(Hint: Connect segment AC which intersects EF at some point G, and find some similar triangle pairs).

